Genetic variation regulation - Quantitative Trait Loci

link DNA-level variation (genome) with RNA-level consequences (transcriptome)

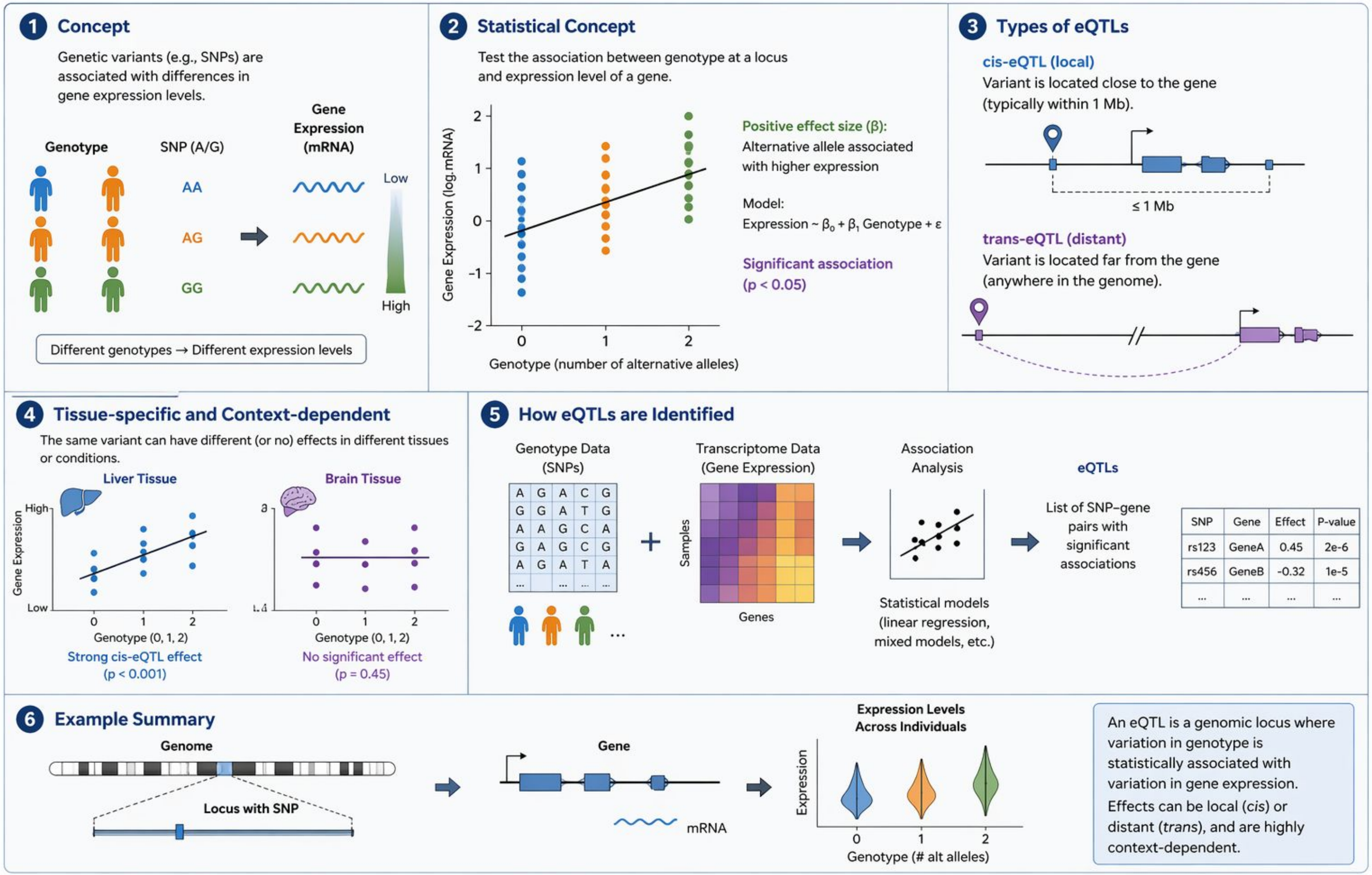
Genomics - transcriptomics integration

- Identify how genetic variation regulates:
  - gene expression (eQTL)
  - splicing (sQTL)
  - allele-specific expression (ASE) - imbalance between alleles
- Connect somatic mutations to transcriptional programs (cancer)
- Connect copy number variation (CNV) to expression variability
- Construct multi-layer gene regulatory networks
- Predict phenotype using combination of gene expressions and variants
- Predict biomarkers and subgroups

Expression Quantitative Trait Loci (eQTL)

- Genomic loci where genetic variation correlates with variation in gene expression levels
- Statistical Concept: Significant association between genotype at a specific locus and the expression level of a gene (typically measured as mRNA abundance)
  - cis-acting (local, typically within 1 Mb of gene)
  - trans-acting (distant, anywhere in genome)
- Tissue-specific and context-dependent
- Identified through correlation of genotypes with transcriptome data

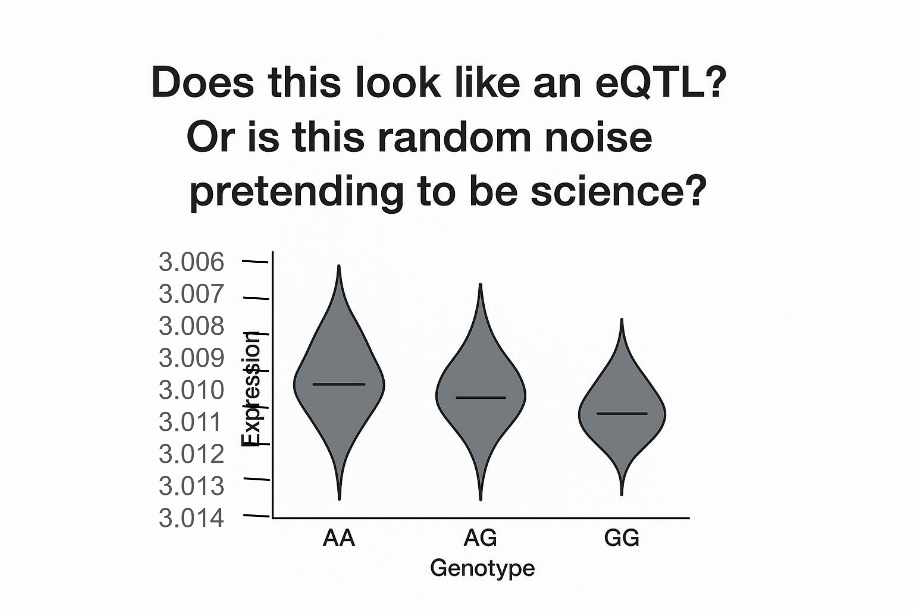
eQTLs: Genomic loci where genetic variation correlates with variation in gene expression



Biological Mechanisms of eQTL Effects

- Biological Mechanisms of eQTL Effects - Transcriptional Regulation
    - Variants in promoter regions affecting transcription factor binding
    - Enhancer variants influencing chromatin accessibility
- Changes in regulatory element activity - post-transcriptional Effects:
    - Variants affecting mRNA stability
    - miRNA binding site alterations
- Structural Variation:
    - Copy number variations affecting gene dosage
    - Chromatin conformation changes

Is this eQTL?



Does this look like an eQTL?
Or is this random noise
pretending to be science?
3.006
3.007
3.008
3.009
3.010
3.011
3.012
3.013
3.014
Expression
AA
AG
GG
Genotype

eQTL Example 1: FTO Obesity Locus

- GWAS identified FTO region variants strongly associated with obesity
- Discovery: Original assumption that FTO gene itself was causal
- eQTL Analysis: Revealed obesity-associated variants function as distant enhancers regulating IRX3 and IRX5 expression, not FTO
- Impact: Redirection of functional studies from FTO to IRX3/IRX5 pathways

The role of the FTO (Fat Mass and Obesity Related) locus in regulating body size and composition

eQTL Example 2: Functional effects of genetic variants in causing rheumatoid arthritis

- rs3128921 eQTL at HLA-DPB2 driving its expression, and correlating with clinical severity
- could potentially be used to stratify patients to more aggressive treatment immediately at diagnosis.

[Expression quantitative trait loci analysis in rheumatoid arthritis identifies tissue specific variants associated with severity and outcome]

sQTL Fundamentals

- Definition: Genetic variants associated with variation in mRNA splicing patterns
- What It Measures: Alternative splicing events, exon inclusion/exclusion ratios, splice junction usage
- Key Distinction from eQTL: sQTLs affect RNA processing rather than overall expression levels
- Detection Methods: RNA-seq analyses quantifying isoform ratios, exon inclusion metrics, splice junction reads

Biological Mechanisms of sQTL Effects

- Splice Site Variants:
  - Alterations in canonical splice donor/acceptor sequences
  - Creation or destruction of splice sites
- Splicing Regulatory Elements:
  - Variants affecting exonic/intronic splicing enhancers
  - Variants affecting exonic/intronic splicing silencers
- Transcription Rate Effects:
  - Promoter variants influencing co-transcriptional splicing
  - Elongation rate changes affecting splice site selection
- RNA Structure Effects:
  - Variants altering RNA secondary structure near splice sites

sQTL Example 1: MAPT in Neurodegeneration

- Gene Context: MAPT encodes tau protein, central to Alzheimer's and other tauopathies
- sQTL Discovery: Multiple variants in MAPT region associated with alternative splicing of exons 2, 3, and 10
- Functional Consequence: Altered 3R/4R tau isoform ratio
- Disease Relevance: Imbalanced 3R/4R ratio contributes to tau aggregation and neurodegeneration

Lineage-specific splicing regulation of MAPT gene in the primate brain

sQTL Example 2: Cancer Susceptibility

- Gene: BRCA2 in breast/ovarian cancer
- sQTL Variant: rs1799954 affects splicing regulatory element
- Mechanism: Alters exon inclusion patterns, producing truncated protein isoforms
- Impact: Reduced DNA repair capacity even in absence of canonical BRCA2 mutations

[Impact of BRCA1 and BRCA2 variants on splicing: clues from an allelic imbalance study]

sQTL Example 3: Pharmacogenomics Application

- CYP3A5 gene expression variation affects metabolism of numerous drugs
- sQTL Discovery: rs776746 (CYP3A5*3) creates alternative splice site leading to premature stop codon
- Functional Impact: Reduced CYP3A5 expression and enzyme activity in *3/*3 homozygotes
- Clinical Relevance: Guides dosing for immunosuppressants, antipsychotics, and opioids

[The Functional Implications of Common Genetic Variation in CYP3A5 and ABCB1 in Human Proximal Tubule Cells]

Allele-Specific Expression (ASE)

- Definition: Imbalanced expression between the two alleles of a gene in diploid individuals
- Detection Requirement: Heterozygous individuals with distinguishable alleles (coding or UTR variants)
- Measurement: Allelic imbalance in RNA-seq reads mapping to each allele
- Key Insight: Reveals cis-regulatory effects in individual samples rather than population averages

ASE Example 1: Autoimmune Regulation

- Gene: IL20RA in psoriasis susceptibility
- ASE Discovery: Heterozygous coding variant rs1558744 shows consistent allelic imbalance
- Mechanism: Variant creates transcription factor binding site differential
- Functional Consequence: Reduced IL20 signaling contributing to epidermal dysregulation

eQTL vs. sQTL vs. ASE: Key Differences

Feature	eQTL	sQTL	ASE
Primary Measure	Total expression level	Splicing patterns/isoform ratios	Allelic expression ratio
Detection Scale	Population-level	Population-level	Individual-level
Typical Analysis	Association testing	Isoform quantification	Allelic read counting
Resolution	Gene-level	Exon/isoform-level	Base-pair level

eQTL vs. sQTL vs. ASE: Key Differences

- eQTL Mapping:
  - Strength: Statistical power from population data
  - Limitation: Cannot distinguish cis from trans without additional experiments
- sQTL Mapping:
  - Strength: Reveals isoform diversity
  - Limitation: Requires deeper RNA-seq and sophisticated quantification
- ASE Analysis:
  - Strength: Within-individual control, sensitive to cis-effects
  - Limitation: Requires heterozygous markers, not all genes analyzable
- Integrated Approaches: Combining methods increases discovery and validation

Data Integration Methodology

- Central Challenge: Most GWAS hits are non-coding with unclear target genes
- Novel Approach: Summary-data-based Mendelian Randomization (SMR)
  - Uses eQTL data as instrumental variables to test causal relationships between gene expression and traits
- Data Requirements: Only summary statistics from GWAS and eQTL studies needed, not individual-level data
- Combine:
  - SNP → expression (eQTL effect)
  - SNP → trait (GWAS effect)
  - to infer:
  - expression → trait (causal link)
- Correlation or causation: MR uses genetic variation as a natural experiment to infer whether a biological factor causally affects a trait.

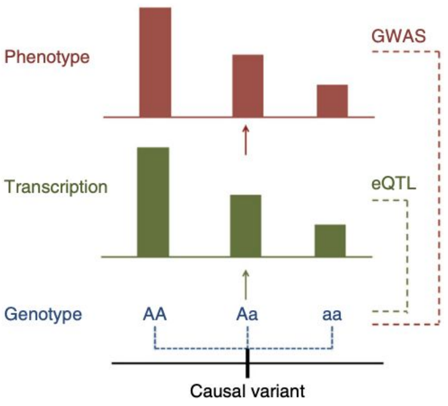
Flagship publication: Integration of summary data from GWAS and eQTL studies predicts complex trait gene targets - Nature genetics (2016)

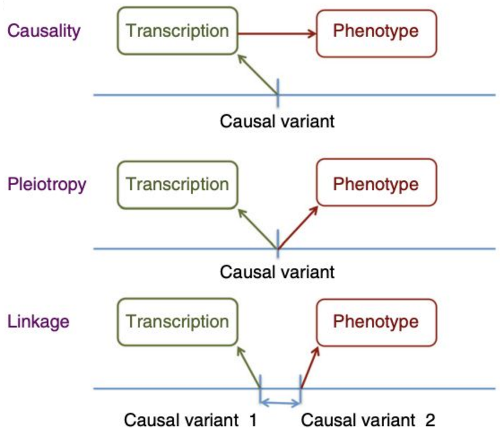
- SMR (Summary-data-based Mendelian Randomization)
- Up to 339,224 individuals across 5 complex traits, 5,311 with eQTL data
- Prioritized 126 genes for 5 complex traits
  - 25 genes were novel candidates not previously identified
  - 77 genes were not the nearest gene to GWAS SNP (challenges "nearest gene" assumption)
  - TRAF1 and ANKRD55 for rheumatoid arthritis
  - SNX19 and NMRAL1 for schizophrenia

Integration of summary data from GWAS and eQTL studies predicts complex trait gene targets

SMR (Summary-data-based Mendelian Randomization)

- Test causal relationships between gene expression and traits
- Only summary statistics from GWAS and eQTL studies needed, not individual-level data
- Identify cis-eQTLs for each gene at genome-wide significance
- Extract summary statistics for these eQTLs from GWAS data
- Perform SMR test to assess association between gene expression and trait
- Apply HEIDI test to distinguish pleiotropy (single gene or genetic variant influences multiple phenotypic traits) from linkage (heterogeneity in dependent instruments)
- Prioritize genes with significant SMR and non-significant HEIDI tests





[Integration of summary data from GWAS and eQTL studies predicts complex trait gene targets]

eQTL Notebook: Linear Regression: The eQTL Workhorse

- The Model:
  - Expression = B0 + B1 x Genotype + error
- What each part means:
  - B0 (intercept) = baseline expression when genotype = 0
  - B1 (slope) = change in expression per allele copy
  - error = everything else affecting expression
- Key outputs:
  - Beta (effect size): how much does expression change?
  - P-value: is this change statistically significant?
  - R-squared: how much variance does the SNP explain?

Multiple Regression: Adding Covariates

- Extended Model:
  - Expression = B0 + B1 x Genotype + G1 x Cov1 + G2 x Cov2 + ... + error
- What changes:
  - B1 is now the SNP effect AFTER accounting for covariates
  - We 'partial out' the covariate effects
- In our analysis:
  - Simple model: 160 significant associations
  - With 10 covariates: number changes! (check notebook)
- Some associations gained, some lost significance
  - Gained = real signals unmasked by removing noise
  - Lost = fake signals that were driven by confounders

PCA: Finding Hidden Confounders

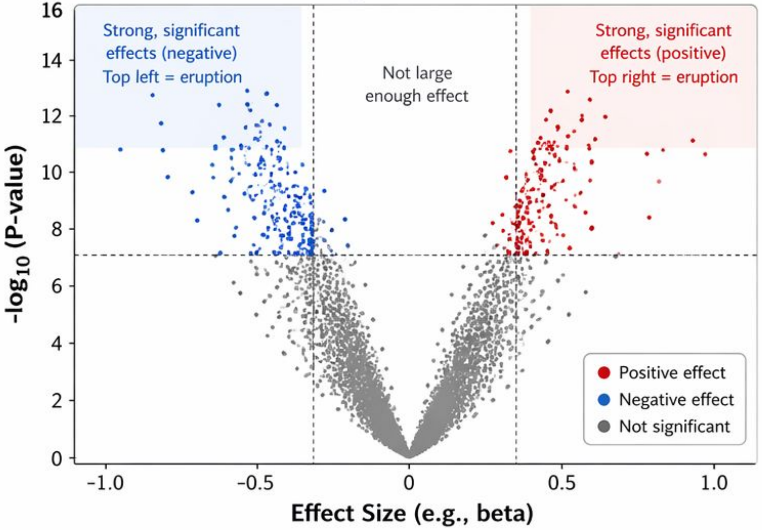
- The Problem:
  - We don't always know all the confounders
  - Unknown batch effects, hidden population structure
- PCA to the rescue:
  - Finds the main axes of variation in the data
  - PC1 = direction of maximum variance (biggest confounder)
  - PC2 = next biggest, orthogonal to PC1, etc.
- Strategy for eQTL analysis:
  - Compute PCs from covariate or expression data
  - Use top PCs as covariates in regression
  - Captures systematic variation without knowing the source

Essential QC Plots Before Interpreting eQTL Results

Two key checks: Are there strong, significant associations? Is the P-value distribution as expected?

1) Volcano Plot (Effect Size vs -log$_{10}$ P-value)

X-axis: how big is the effect?
Y-axis: how significant is it?

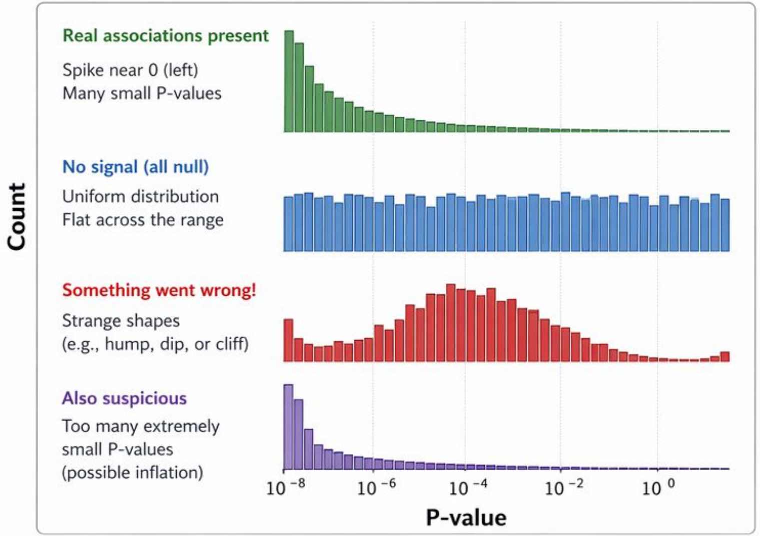


What to look for:

- Many points at the top corners (red/blue) = strong, significant eQTLs
- Symmetry around 0 (mostly) is typical
- Lack of high points = little power or no real signal

2) P-value Distribution (Histogram)

What does the distribution of P-values look like?



What to look for:

- Spike near 0 = real associations
- Uniform = no signal (as expected under all null)
- Strange shapes = potential issues (e.g., QC problems, batch effects, population structure, model misspecification)

Why both plots matter: The volcano plot shows the magnitude and significance of effects. The P-value histogram checks whether the overall pattern of significance makes sense.

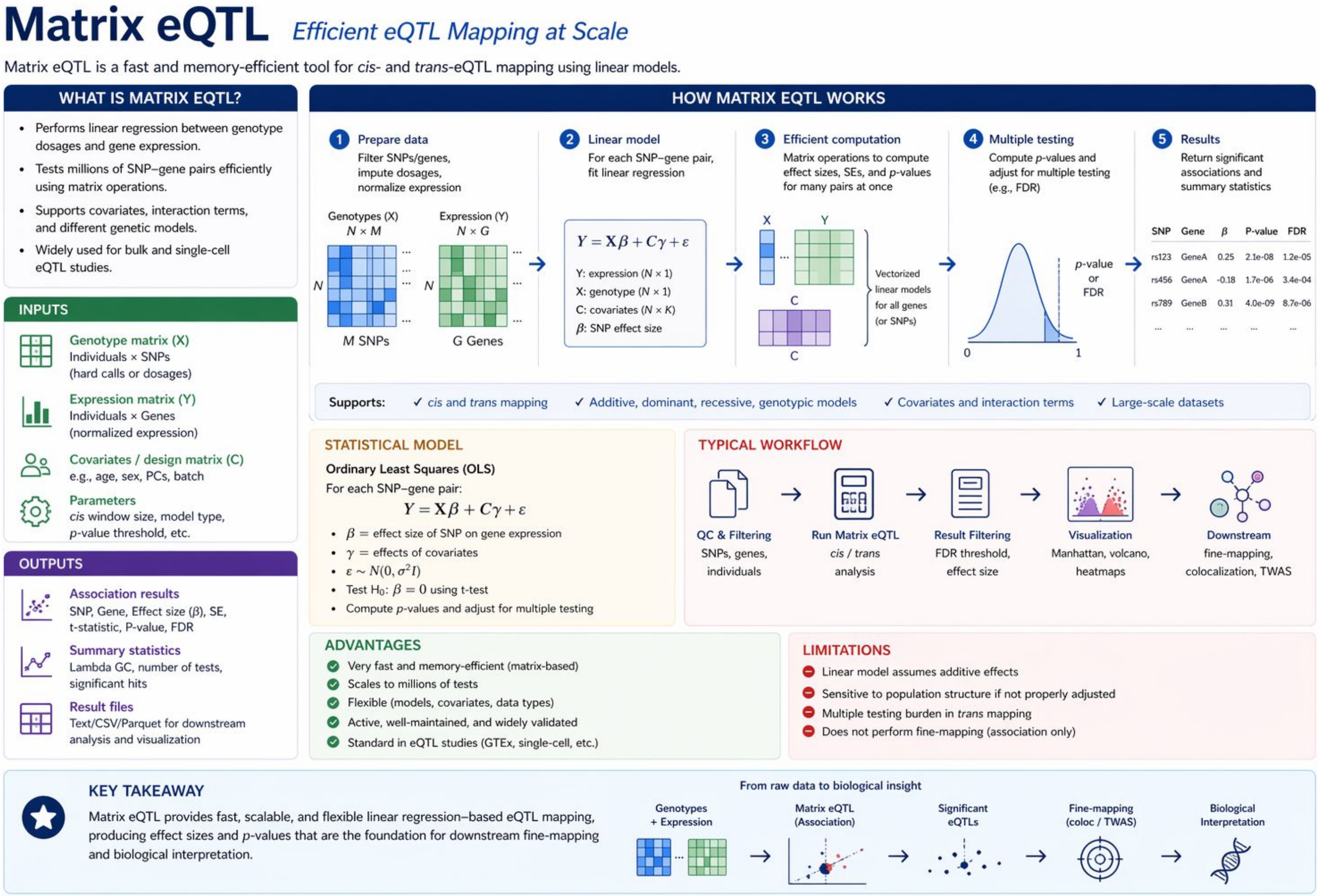
→ **Do these QC checks before trusting or interpreting your eQTL results!**

Understanding Effect Sizes and Confidence Intervals

- Effect Size (Beta)
  - Positive beta: more alt alleles = higher expression
  - Negative beta: more alt alleles = lower expression
  - Units match the expression data units
- 95% Confidence Interval
  - Range where the true effect likely falls
  - If CI crosses zero: effect might be zero (not significant)
- Example: snp_4 vs CACNA1B
  - Beta = -6.98, CI: [-10.95, -3.01], p = 0.0006
  - CI does NOT cross zero - significant negative effect!

Three Models Compared

- Model 1: Simple (Expression ~ Genotype)
  - Baseline - no correction for confounders
- Model 2: With Covariates (+ known covariates)
  - Better - accounts for known confounding variables
- Model 3: With PCs (+ principal components)
  - Best - captures known AND unknown confounders
- What we see:
  - R-squared generally improves with more covariates
  - Number of significant eQTLs changes across models
  - Top hits tend to be stable across all three models



Matrix eQTL tests each SNP independently and does not model Linkage Disequilibrium (LD) or multiple SNP effects jointly.

MatrixEQTL: The Industry Standard

- **MatrixEQTL (Shabalin 2012, Bioinformatics)**
  - Ultra-fast eQTL analysis via matrix operations
  - Tests millions of SNP-gene pairs in minutes
- **The Model (linear regression with covariates):**
  - Expression = B0 + B1*Genotype + B2*Age + B3*Sex + B4*PC1 + ... + error
  - Tests H0: B1 = 0 (no effect of genotype on expression)
- **Key features:**
  - Cis/trans distinction using genomic coordinates
  - Covariate correction (confounders removed)
  - FDR multiple testing correction (Benjamini-Hochberg)
- **Used by: GTEx, ENCODE, eQTL Catalogue, UK Biobank**

Covariate Preparation & R Script

- **Covariates constructed from data:**
    - Age and Sex from ASD_covariates.csv
    - PC1, PC2, PC3 from genotype PCA (population structure)
    - 5 covariates total, transposed for MatrixEQTL format
- **R script (run_matrixeqtl.R) configures:**
    - Model: linear regression (modelLINEAR)
    - Cis distance: 1 Mb (1e6 bp)
    - P-value thresholds: output ALL results (pvThreshold = 1.0)
    - FDR correction applied across all tests
- **Docker image: pl92297/matrixeqtl**
    - Pre-built image with R + MatrixEQTL package

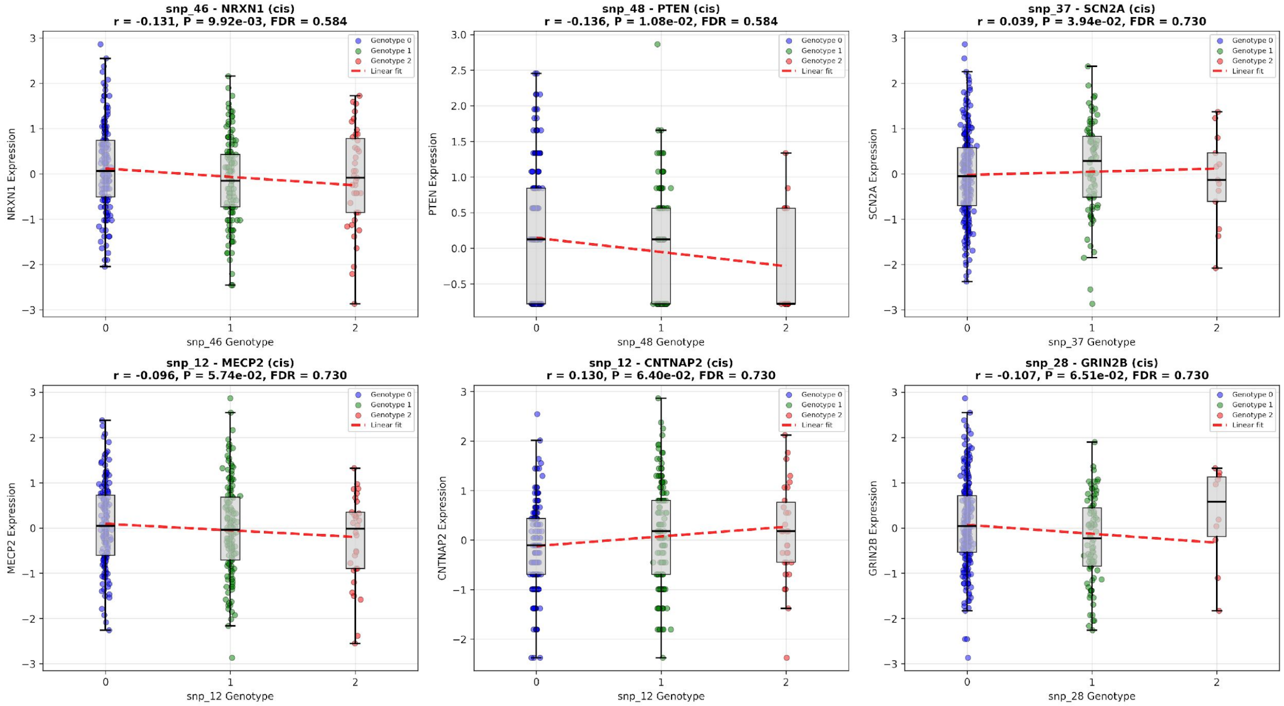
Running MatrixEQTL in Docker

- **Docker command:**
  - docker run --rm -v {data_dir}:/data pl92297/matrixeqtl Rscript /data/run_matrixeqtl.R
- **What MatrixEQTL does internally:**
  - Loads expression, genotype, and covariate matrices
  - Loads gene and SNP position files
  - For each SNP-gene pair within cis distance:
  - Fits: expression ~ genotype + covariates
  - Computes t-statistic and p-value
- **Outputs:**
  - 108 cis SNP-gene tests (out of 500 possible pairs)
  - Each with: SNP, Gene, t-statistic, p-value, FDR, beta, type

Understanding the Top Hits

- **snp_46 -> NRXN1: best association (P = 0.010)**
  - $t = -2.60$, $r = -0.149$ (negative effect) ($t = \beta / SE(\beta)$)
  - After INT, effect sizes are standardized and comparable across genes
- **snp_48 -> PTEN: second best (P = 0.011)**
  - $t = -2.57$, $r = -0.147$ (negative effect)
  - PTEN and NRXN1 are both ASD-associated genes
- **Only 2 pairs reach nominal significance (P < 0.05)**
  - With 108 tests, we expect ~5 by chance alone
  - 2 is below expectation - INT removed count-scale artifacts
- **FDR correction is conservative but necessary**

Top eQTL Associations (Scatter/Box Plots)



What it shows

Top 6 SNP-gene pairs by p-value. Each panel: expression (Y) vs genotype (X = 0, 1, 2).

Box + scatter overlay

Box plots summarize distribution per genotype group. Jittered dots show individual samples (blue=0, green=1, red=2).

Red dashed line

Linear regression fit. Slope = beta (effect size). Steeper = stronger eQTL.

Top-left: snp_46-NRXN1

Best hit (P=0.01). Negative slope: minor allele decreases NRXN1 expression. r=-0.149.

Top-right: snp_48-PTEN

Second best (P=0.011). Negative slope. PTEN is a known ASD risk gene.

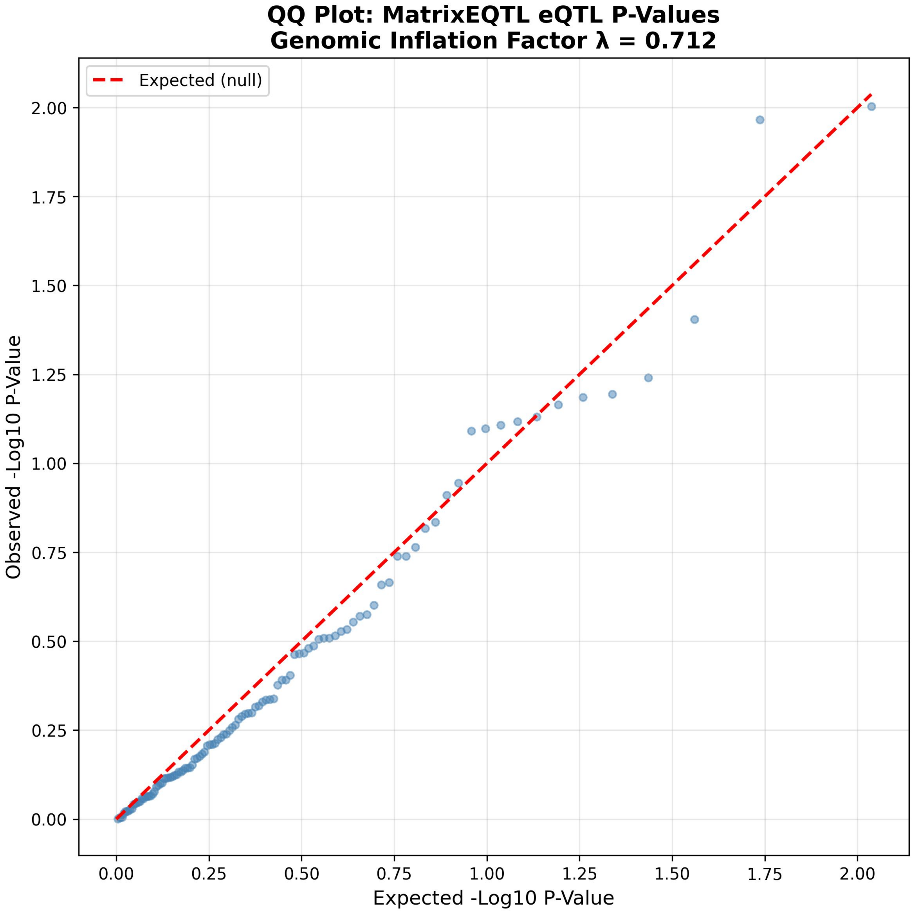
Y-axis is normalized

After INT, expression ranges -3 to 3 (standard normal). Effect sizes are comparable across all genes.

All FDR > 0.58

None pass FDR correction. The trends are suggestive but not statistically reliable after multiple testing.

QQ Plot of P-Values



What it shows
Quantile-quantile plot comparing observed p-values (Y) to expected under the null hypothesis (X). Both as -log10.

Diagonal red line
Expected distribution if NO eQTLs exist (uniform p-values). Points on this line = no signal.

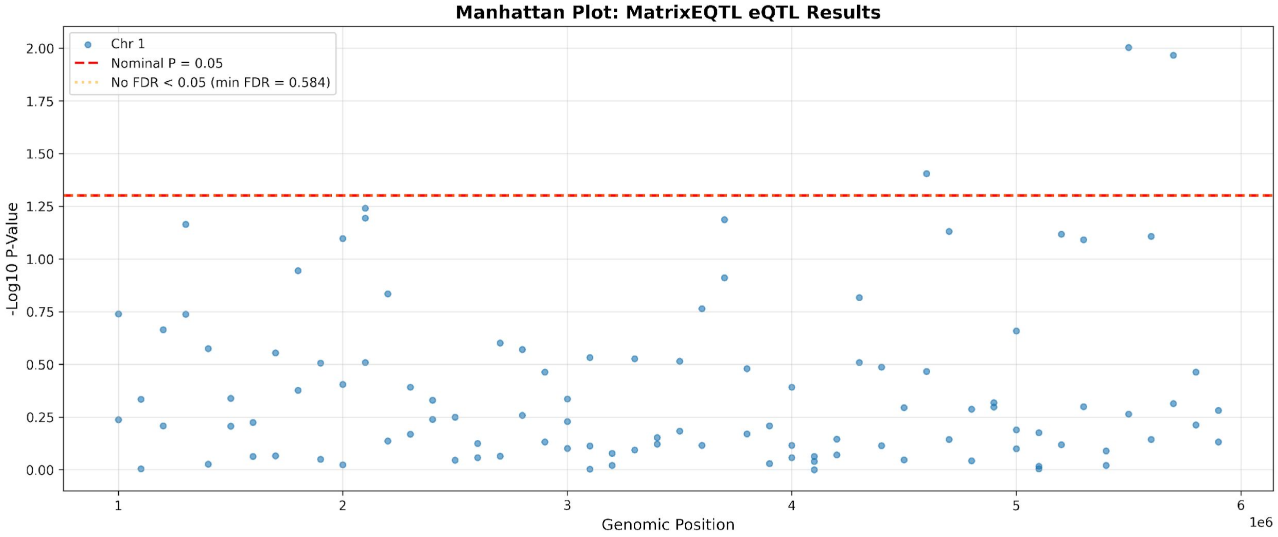
Deviation upward
Points above the diagonal at the right tail indicate more significant p-values than expected by chance.

Genomic inflation factor
Lambda = 0.712. Values near 1.0 = well-calibrated. Ours is slightly deflated after INT normalization.

Deflation meaning
Lambda < 1 means fewer significant p-values than expected by chance. INT removed count-scale artifacts that previously inflated results.

Real-data comparison
Strong eQTL datasets show dramatic upward curves. Our flat/deflated QQ is typical of small datasets with weak effects after proper normalization.

Manhattan Plot



What it shows

Each dot is a SNP-gene test. X = genomic position. Y = -log10(P-value). Higher = more significant.

Red dashed line

Nominal significance threshold (P = 0.05, -log10 = 1.3). 12 dots rise above this line.

Orange dotted line

FDR = 0.05 threshold. Label says 'No FDR < 0.05 (min FDR = 0.584)' - nothing passes.

Highest dots

snp_46-NRXN1 and snp_48-PTEN at -log10(P) ~2.0. The only two above the nominal line.

Scattered pattern

No tight peak clusters (unlike real GWAS). Synthetic positions spread tests across chr1.

All tests are cis

108 cis tests plotted. No trans tests because all SNPs/genes are on synthetic chr1 within 1 Mb.

What MatrixEQTL Tells Us (and Doesn't)

- **What it tells us:**
  - Which SNP-gene pairs show marginal association
  - P-values quantify evidence against the null (no effect)
  - Effect sizes (beta) quantify the direction and magnitude
  - FDR controls the rate of false discoveries
- **What it does NOT tell us:**
  - Which SNP is actually CAUSAL (LD confounds this)
  - Whether the association is mediated or direct
  - What biological mechanism underlies the eQTL
- **For causality: need fine-mapping (SuSiE, FINEMAP)**
- **For mechanism: need functional experiments (CRISPR)**

Output Files Reference

- **CSV Data File:**
  - Phase2_MatrixEQTL_results.csv - All 108 cis eQTL tests
  - Columns: SNP, Gene, Statistic, P_Value, FDR, beta, Type, Significant
- **Visualization Files:**
  - Phase2_MatrixEQTL_top_associations.png - Top 6 eQTL scatter plots
  - Phase2_MatrixEQTL_qq_plot.png - QQ plot with lambda
  - Phase2_MatrixEQTL_manhattan_plot.png - Manhattan plot
- **Supporting Files:**
  - scripts/MatrixEQTL.py - Full pipeline script
  - data/ASD_dataset/run_matrixeqtl.R - Generated R script
- **Docker image: pl92297/matrixeqtl**

Genotype Dosage

- **Definition:** Expected number of copies of an allele (continuous value between 0 and 2)
- **Hard genotype vs dosage:**
  - Hard calls: **0 / 1 / 2** (discrete)
  - Dosage: **real-valued** (e.g., 0.3, 1.2, 1.8)
- **Computation:** $\text{dosage} = 0 \cdot P(0) + 1 \cdot P(1) + 2 \cdot P(2)$
- **Why it is used:**
  - Captures **uncertainty** from imputation
  - Preserves more information than hard calls
  - Improves statistical power especially on low-coverate samples
- **In eQTL models:**
  - Used as a **continuous predictor**
  - Example: $Y = \beta \cdot \text{dosage} + \varepsilon$
- **Key takeaway:**
  Dosage provides a probabilistic, information-rich representation of genotype

How to calculate Genotype Dosage?

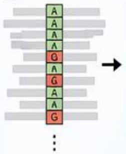
DS is just a number in [0, 2] — the source depends on the upstream pipeline.
There are **three main origins:**

1 From reads directly (sequencing-based calling)

When you run a variant caller like GATK HaplotypeCaller, DeepVariant, bcftools call, or DRAGEN on your own BAM/CRAM files, DS **(if present)** is computed from your sample's reads at that position.

Read pileup at a SNP

Ref A



(example: A ↔ G)

→ Caller computes genotype likelihoods: P(reads | AA), P(reads | AB), P(reads | BB)

→ Apply prior (flat or HW-based) → posteriors: P(AA | reads), P(AB | reads), P(BB | reads)

→ Genotype Dosage (Expected # of alt alleles): DS = P(AB | reads) + 2 · P(BB | reads)

$0 \le DS \le 2$

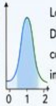
- This is exactly the Bayesian calculation
- No external database involved — the evidence comes entirely **from reads at that locus in that sample.**

Typical context:

WGS, WES, low-coverage sequencing (lcWGS)

High-coverage WGS (>30×): DS is nearly always 0, 1, or 2 because read evidence is overwhelming.

Low coverage (1–4×): DS becomes genuinely continuous and informative.



2 From imputation (reference-panel-based)

This is where public databases enter the picture. Imputation fills in variants you didn't directly genotype/sequence by leveraging linkage disequilibrium (LD) patterns from a reference panel.

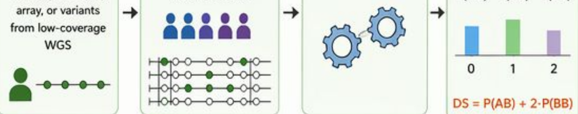
How it works

Your sparse genotypes: e.g., 500K SNPs from a genotyping array, or variants from low-coverage WGS

→ Reference panel (public database): e.g., 1000 Genomes, HRC, TOPMed

→ Imputation tools (Beagle, Minimac4, IMPUTE5)

→ Imputed variants (posterior genotype probabilities → DS)



DS = P(AB) + 2·P(BB)

Here, DS at an imputed site reflects:

- The observed genotypes at nearby typed sites (your data)
- Linkage patterns learned from the reference panel (public database)

Typical scenario

Input: sample.vcf.gz with 500K genotyped SNPs (GT only)
Run: TOPMed Imputation Server (or Michigan Imputation Server)
Output: sample.imputed.vcf.gz with ~40M variants, each with GT, DS, GP, and INFO/R^2

★ **The R^2 (or INFO score)** tells you how confidently the variant was imputed — values near 1 mean "nearly as good as directly typed," values near 0 mean "essentially a guess from allele frequency."

3 Hybrid: genotype refinement

Some pipelines use reads plus a reference panel to refine calls. Tools like **GLIMPSE** (for low-coverage WGS) or **BEAGLE 5** with a reference panel combine:

Read-based genotype likelihoods (from your sample): P(reads | AA), P(reads | AB), P(reads | BB)

+

LD information (from the reference panel)

↓

Model combines both sources of evidence → Output DS reflects both sources of evidence

$0 \le DS \le 2$

★ This gives more accurate dosages than either source alone, especially at low coverage.

Key takeaway: Genotype Dosage (DS) is the expected number of alternate alleles (0 to 2). It can come from reads directly, from reference-panel imputation, or from a hybrid approach that combines both. The confidence in DS depends on the amount and quality of evidence behind it.

Linkage Disequilibrium (LD)

- **Definition:** Non-random association of alleles at different loci
- **Key idea:** Nearby SNPs are **correlated** due to shared inheritance
- **Mathematically:**
  $P(AB) \neq P(A) \times P(B)$
- **Main causes:**
  - Physical proximity on chromosome
  - Limited recombination
  - Population history (selection, bottlenecks)
- **Common metrics:**
  - **Pearson** correlation strength (most used)
  - **D′** → recombination history
- **Why LD matters:**
  - Creates **clusters of associated SNPs** in GWAS/eQTL
  - Top SNP ≠ necessarily causal
  - Requires **fine-mapping** (e.g., SuSiE, FINEMAP)

LD Problem

- **Association != Causation (especially in genetics)**
  - Nearby SNPs are correlated due to Linkage Disequilibrium (LD)
  - If causal SNP A is in LD with SNPs B, C, D - all four look significant
- **MatrixEQTL finds associations - but can't resolve LD**
  - Our results: 12 nominally significant pairs, 0 FDR-significant
  - Many of these are likely the same signal seen through correlated SNPs
- **We need a method that models LD explicitly**
  - Fine-mapping asks: 'Given the correlation structure, which SNP is most likely causal?'

SuSiE (Sum of Single Effects)

Scalable Bayesian Fine-Mapping

HOW IT WORKS (INTUITION)

- Models the phenotype as a sum of sparse single-effect components
- Iteratively fits one signal at a time using variational inference
- Each component explains one causal variant (or signal)
- Naturally handles multiple causal variants in a locus
- Performs Bayesian variable selection while accounting for LD

INPUTS

Genotype data (X)
SNPs × individuals or LD matrix (**R**)

Phenotype (y) or summary statistics (z)

Number of effects (L)
Upper bound on the number of causal variants

Other options
e.g., prior variance, standardization

OUTPUTS

Posterior Inclusion Probability (PIP) to phenotype
Probability each SNP is causal

Credible sets
Set of variants likely to contain the causal variant for each signal

Effect size estimates (β)
Per signal and per SNP

HOW Susie WORKS – STEP BY STEP

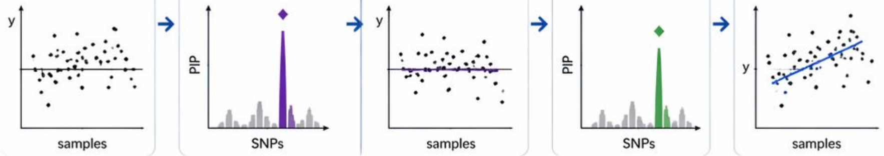
1 Initialize
Residual $r^0 = y$ (nothing explained yet)

2 Fit one effect ($l = 1$)
Find the SNP most likely to be causal for the current residual

3 Update
Estimate effect size β^1 and update residual $r^1 = y - X\beta^1$

4 Repeat for $l = 2 \dots L$
Find next signal in the new residual r^{l-1}

5 Final model
Sum of L single effects explains the data



Remove explained signal → search for additional independent signals

MODEL

SuSiE decomposes the genetic effect into L single-effect components

$$y = X\beta + \varepsilon$$

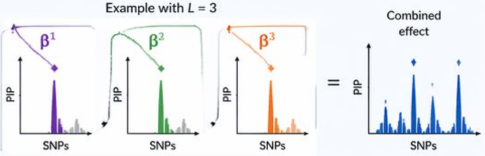
Phenotype = Genotypes × Effects + Noise

SuSiE:

$$y = \sum_{l=1}^{L} X\beta^{(l)} + \varepsilon$$

Each $\beta^{(l)}$ is sparse: only one (or few) SNPs have non-zero effects

Example with $L = 3$



ADVANTAGES

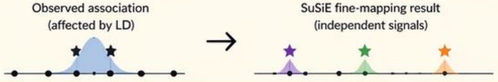
- Scales well to large datasets (fast and stable)
- Handles high LD regions robustly
- Allows multiple causal variants (multiple signals)
- Produces interpretable credible sets
- Works with individual-level data or summary statistics

LIMITATIONS

- Requires choosing the number of effects (L)
- Uses approximate variational inference (not exact)
- Highly correlated signals can merge (if two causal variants are extremely correlated)

KEY TAKEAWAY

SuSiE transforms a cluster of correlated SNP associations into a small number of independent signals, providing probabilistic causal hypotheses (PIPs and credible sets).



What is Susie

- **SuSiE = Sum of Single Effects (Wang et al. 2020, JRSS-B)**
  - A Bayesian variable selection method for fine-mapping
- **The Model:**
  - y = X*b1 + X*b2 + ... + X*bL + error
  - Each bl has exactly ONE non-zero entry (one causal SNP)
  - L = max number of independent causal signals
- **Key Outputs:**
  - PIP: Posterior Inclusion Probability per SNP
  - Credible Sets: groups of SNPs likely containing the causal variant
  - Effect sizes: estimated effect of each SNP on expression

L Single-Effect Components

- **Each component l (l = 1..L) represents one causal signal**
  - y = X*b1 + X*b2 + ... + X*b5 + error (L=5 in our analysis)
- **Within each component:**
  - Exactly ONE SNP has a non-zero effect
  - SuSiE doesn't know which - it assigns a probability to each
  - These probabilities form a distribution over all 50 SNPs
- **PIP aggregates across all components:**
  - PIP(SNP_j) = Prob(SNP_j is causal in at least one component)
- **Empty components get pruned automatically**
  - L=5 doesn't mean 5 causal SNPs - unused components shrink away

PIP vs Effect: Two Different Questions

- **PIP = 'How confident are we this SNP is causal?'**
  - Probability that SNP has non-zero effect in any component
  - Range: 0 (definitely not causal) to 1 (definitely causal)
- **Effect = 'If it IS causal, how big is the impact?'**
  - Posterior mean effect size (change in expression per allele)
  - Averaged across all L components, weighted by probability
- **They are independent:**
  - High PIP + Small Effect = reliable but weak regulatory variant
  - Low PIP + Large Effect = uncertain but potentially important
  - High PIP + Large Effect = strong fine-mapping hit

R Script & Docker Execution

- **R Script (run_susie.R):**
  - Loads genotype matrix X (300 x 50, coerced to double)
  - For each gene: fit = susie(X, y, L=5)
- **Key SuSiE Parameters:**
  - L = 5: max number of causal signals per gene
  - scaled_prior_variance = 0.2: fixed prior (20% of var(y))
  - estimate_prior_variance = FALSE: prevents shrinkage to zero
  - coverage = 0.5: credible set coverage threshold
- **Docker Image: eqtlcatalogue/susie-finemapping**
  - Contains R + susieR package (used by eQTL Catalogue project)
  - Fallback: local Rscript if Docker unavailable

Phase2_SuSiE_gene_summary.csv

- Per-gene overview of fine-mapping results
  - Gene: gene identifier (gene_1 through gene_10)
  - N_Credible_Sets: how many independent causal signals found
  - Top_SNP: SNP with the highest PIP for that gene
  - Top_PIP: the highest PIP value
  - Converged: whether the SuSiE algorithm converged
- Key findings:
  - All 10 genes converged successfully
  - All 10 genes have at least 1 credible set; gene_2 has 2
  - Top PIP: 0.394 (snp_44 for gene_9) - moderate confidence
  - No PIP > 0.5 - consistent with weak signals in this dataset

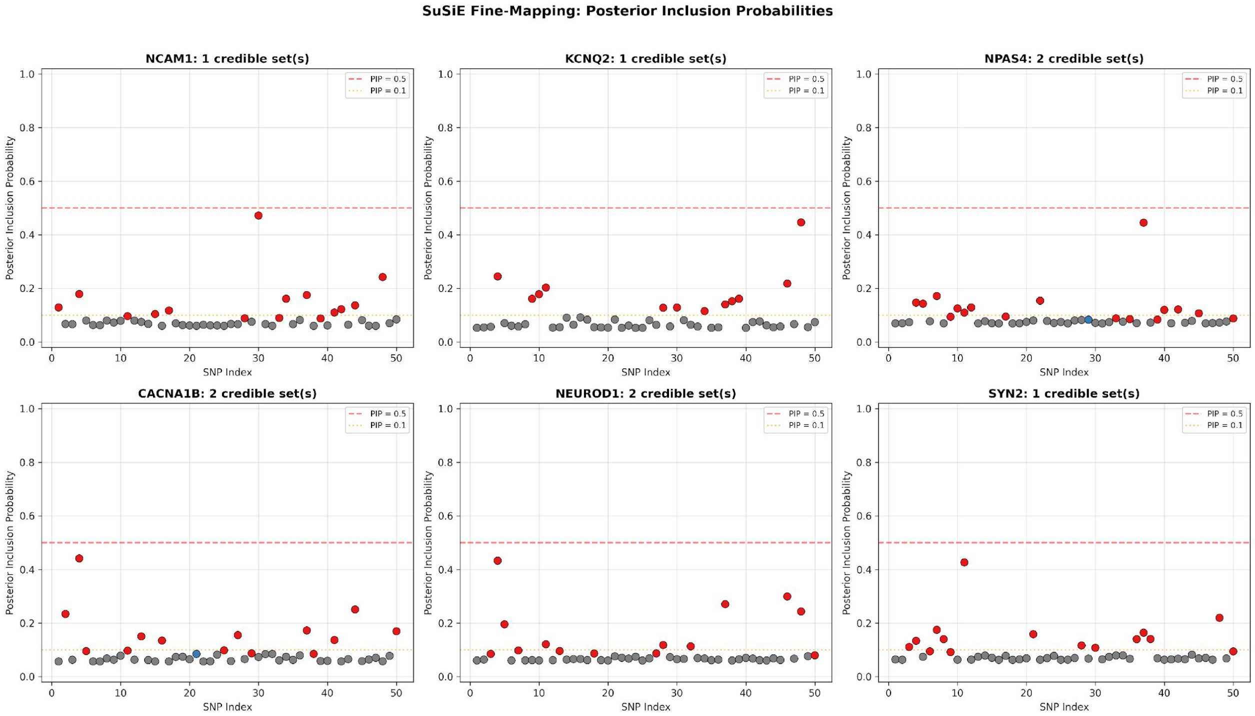
Phase2_SuSiE_results.csv

- **Full results: 500 rows (10 genes x 50 SNPs)**
  - Gene, SNP: the SNP-gene pair
  - PIP: posterior inclusion probability (0 to 1)
  - Effect: posterior mean effect on expression per allele
  - CS_ID: credible set membership (NA if not in any set)
- **How to read it:**
  - Sort by PIP descending to find the most likely causal SNPs
  - Filter CS_ID != NA to see only credible set members
- **Example row:**
  - gene_9, snp_44, PIP=0.394, Effect=-5.04, CS_ID=1
  - "39% chance snp_44 is causal, reducing expression by ~5 units"

Phase2_SuSiE_credible_sets.csv

- **Only SNPs that belong to a credible set (184 rows)**
  - Gene, CS_ID: which gene and which credible set
  - SNP, PIP, Effect: same as results table
  - CS_Size: number of SNPs in this credible set
  - CS_Coverage: cumulative PIP of the set
  - CS_Min_Corr: minimum pairwise |correlation| in the set
- **Example: gene_1, CS_ID=1**
  - 14 SNPs in the credible set, coverage = 50.2%
  - Top SNP: snp_44 (PIP=0.30), then snp_15 (PIP=0.26)
- **Interpretation:**
  - Large CS (14-18 SNPs): weak signal, poor resolution
  - In real data with strong signals: CS sizes of 1-5 are common

PIP Manhattan Plot



What it shows

Each panel is one gene. Y-axis is the Posterior Inclusion Probability (0-1) for each SNP.

Red vs gray dots

Red = SNP belongs to a credible set (causal candidate). Gray = not in any set.

Dashed lines

Red dashed = PIP 0.5 threshold. Orange dashed = PIP 0.1 threshold.

Best signal

NCAM1: snp_30 reaches PIP~0.47 - the highest in our dataset. No SNP crosses 0.5.

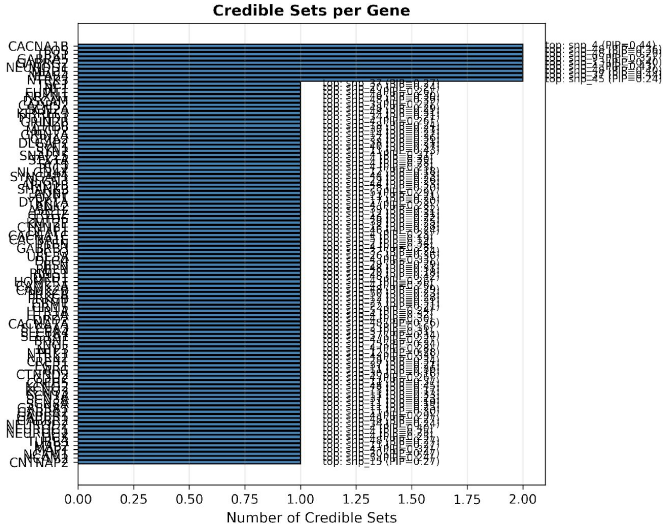
Weak signal pattern

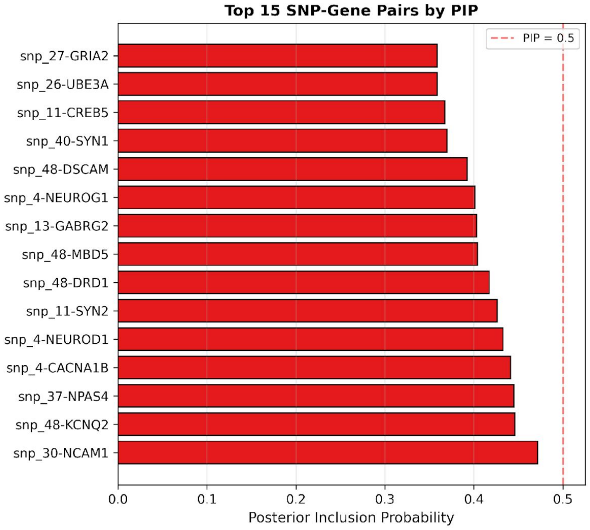
Many SNPs have similar moderate PIPs (0.1-0.4). SuSiE is honestly uncertain - consistent with no FDR-significant eQTLs in MatrixEQTL.

In real data

Strong eQTLs produce 'lollipop' patterns: one SNP at PIP~1.0, all others near 0.

Credible Set Summary





Left panel: CS per gene

Blue bars show number of credible sets. 9 genes have 2 sets (e.g., NPAS4, CACNA1B, NEUROD1).

Annotations

Each bar is labeled with the top SNP and its PIP. Best: snp_30 (PIP=0.47) for NCAM1.

Right panel: Top 15 by PIP

Ranked SNP-gene pairs. snp_30-NCAM1 leads. The dashed line at PIP=0.5 shows none cross it.

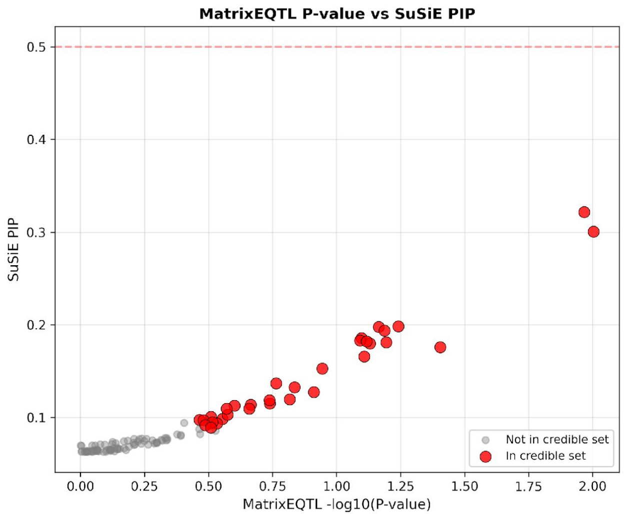
snp_44 appears often

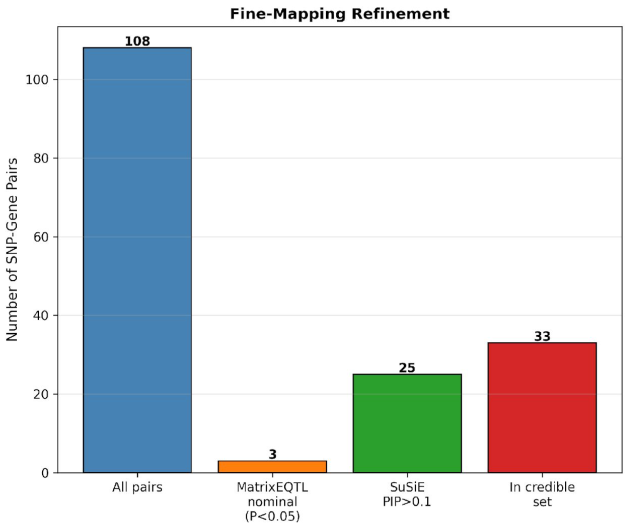
snp_48 appears for multiple genes - but these are separate SuSiE runs, not pleiotropy claims.

9 genes with 2 CS

9 genes have 2 credible sets = SuSiE found 2 independent regulatory signals for each.

SuSiE vs MatrixEQTL Comparison





Left: Scatter plot

X = MatrixEQTL -log10(P). Y = SuSiE PIP. Positive trend: lower p-values tend to have higher PIPs.

Red vs gray dots

Red = in a SuSiE credible set. Gray = not. Credible set members cluster toward higher PIP and lower p-value.

Imperfect correlation

Some SNPs have low p but low PIP (LD-inflated). Others have moderate PIP despite modest p (SuSiE models LD jointly).

Right: Refinement bar chart

108 cis pairs -> 3 MatrixEQTL nominal ($P<0.05$) -> 25 PIP>0.1 -> 33 in credible sets.

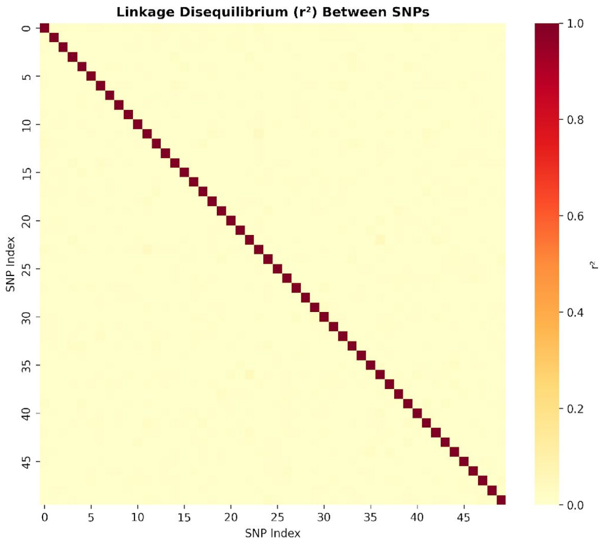
3/3 overlap

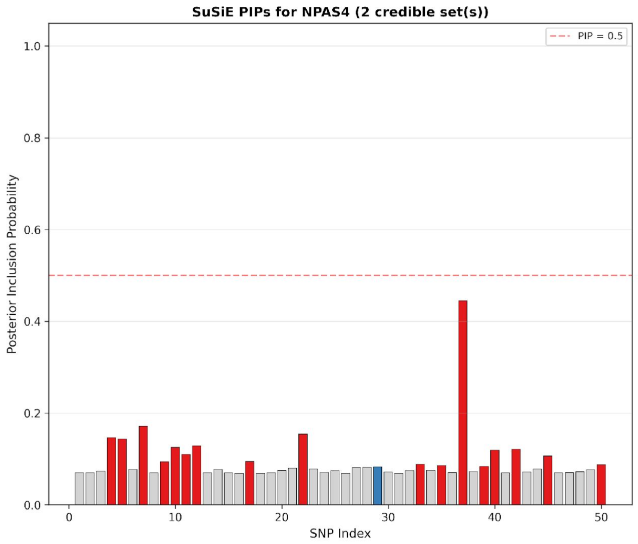
All 3 MatrixEQTL nominal hits are in SuSiE credible sets - perfect concordance between methods.

Key difference

MatrixEQTL asks 'is it associated?' SuSiE asks 'is it causal?' - different questions, complementary answers.

LD Structure and Credible Sets





Left: LD heatmap

r-squared between all 50 SNPs. Dark red on diagonal = self-correlation. Off-diagonal is nearly all yellow = very low LD.

Low LD = poor resolution

When SNPs are uncorrelated, each is independently plausible as causal. SuSiE can't narrow down without LD structure.

Right: NPAS4 PIPs

One of 9 genes with 2 credible sets. Red bars = CS 1. Blue bars = CS 2 only. Gray = not in any CS.

Large credible sets

CS sizes of 14-18 SNPs reflect the weak signal. In real data with LD blocks, CS sizes of 1-3 are common.

Real-world contrast

Human LD blocks let SuSiE say 'these correlated SNPs share one signal - only one needs to be in the CS.' That's how fine-mapping achieves high resolution.

Concrete Examples from Our Results

- **snp_30 -> NCAM1: PIP=0.472, Effect=0.074**
  - Best fine-mapping hit in our dataset
  - 47% chance snp_30 is causal for NCAM1 (effect=0.074 on normalized scale)
  - Moderate confidence, small standardized effect
- **snp_4 -> CACNA1B: PIP=0.441, Effect=-0.078**
  - 44% confidence, comparable standardized effect to NCAM1
  - After INT, effect sizes are on the same scale for all genes
- **snp_48 -> KCNQ2: PIP=0.446, Effect=-0.064**
  - 45% chance of being causal, with small standardized effect
  - Effect sizes are comparable across genes after INT normalization
- **Takeaway: always report PIP AND effect together**

SuSiE Fine-Mapping: Key Takeaways

- **1. SuSiE answers: 'Which specific variants are most likely causal?'**
  - Complements MatrixEQTL (discovery) with fine-mapping (resolution)
- **2. Credible sets are the core output**
  - Smallest set of SNPs likely containing the causal variant
  - Size reflects fine-mapping resolution (smaller = better)
- **3. PIP != Effect - they measure different things**
  - PIP = confidence in causality; Effect = magnitude of impact
- **4. Fine-mapping quality depends on signal strength and LD**
  - Strong signals + LD structure -> precise fine-mapping (CS size 1-3)
  - Weak signals + low LD -> diffuse results (our dataset)
- **5. The pipeline: discover (MatrixEQTL) -> fine-map (SuSiE) -> validate**

Output Files Reference

- **CSV Data Files:**
  - Phase2_SuSiE_results.csv - All 5000 SNP-gene pairs with PIPs
  - Phase2_SuSiE_credible_sets.csv - 1848 SNPs in credible sets
  - Phase2_SuSiE_gene_summary.csv - Per-gene summary (100 rows)
- **Visualization Files:**
  - Phase2_SuSiE_pip_manhattan.png - PIP per SNP per gene
  - Phase2_SuSiE_credible_sets.png - CS count + top PIPs
  - Phase2_SuSiE_vs_MatrixEQTL.png - Method comparison
  - Phase2_SuSiE_ld_credible_sets.png - LD heatmap + CS overlay
- **Script:**
  - scripts/susie.py - Full pipeline (Python + R via Docker)

Summary and Key Takeaways

- eQTLs link genetic variants to gene expression levels, identifying regulatory elements
- sQTLs connect variants to splicing variation, affecting protein isoform production
- ASE reveals allele-specific regulation within individuals with exquisite sensitivity
- Integration methods like SMR connect regulatory variation to complex traits
- Together, these approaches transform GWAS associations into biological understanding
- Run eQTL tutorial and MatrixeQTL